

**Electric potential:** The work done in carrying a unit positive charge from infinity to any point in an electric field is called electric potential at that point in the electric field. The electric potential difference is defined as the work done per unit charge while moving the charge between two points in an electric field.

Suppose we have two points  $A$  and  $B$  in an electric field, move a test charge  $q_0$  from  $A$  to  $B$  and if the work done by the agent moving the charge is  $W_{AB}$ , then electric potential difference is defined as,

$$V_B - V_A = \frac{W_{AB}}{q_0}$$

The work done  $W_{AB}$  may be (i) positive (ii) negative or (iii) zero, so  $V_B$  the potential at  $B$  will be (i) higher (ii) lower or (iii) the same as  $V_A$  respectively.

Generally point  $A$  is considered to be at infinite distance and potential there  $V_A$  is taken as zero, i.e.,  $V_A = 0$ , then potential  $V$  at  $B$  is given by,

$$V = \frac{W_{AB}}{q_0}$$

Thus we define the electric potential at a point as the work done in moving the test charge  $q_0$  from infinite to that point.

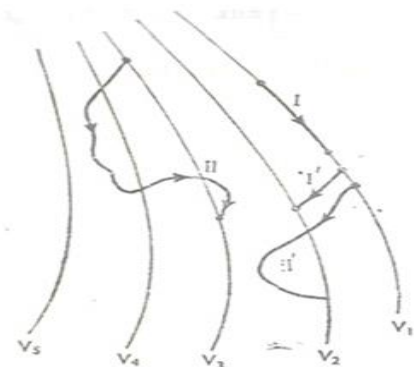
**Electric potential energy:** The electric potential energy of a system of point charges is defined as the work required to assemble this system of charges by bringing them close together, as in the system from an infinite distance.

**Difference between electric potential and electric potential energy:** The differences between electric potential and electric potential energy are:

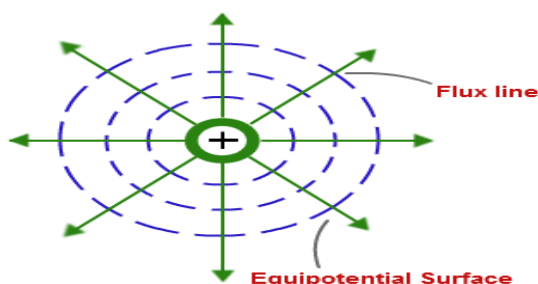
| <b><i>Electric potential</i></b>                                                                                                                                   | <b><i>Electric potential energy</i></b>                                                                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The work done in carrying a unit positive charge from infinity to any point in an electric field is called electric potential at that point in the electric field. | The electric potential energy of a system of point charges is defined as the work required to assemble this system of charges by bringing them close together, as in the system from an infinite distance. |
| It is the property of an electric field, regardless of whether a charged object has been placed in that field.                                                     | It is the energy of a charged object in an electric field                                                                                                                                                  |
| For a point charge the electric potential is<br>$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$                                                                         | For a system of charges electric potential energy is<br>$U = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r}$                                                                                                   |
| Its unit is Joule/Coulomb (J/C) or Volt (V)                                                                                                                        | Its unit is Joule (J)                                                                                                                                                                                      |

### Equipotential surface

The locus of points, all of which have the same electric potential is called an *equipotential surface*. In other words, all points on that surface are at the same potential, such a surface is called an equipotential surface. Fig. 1. shows a few equipotential surfaces. No work is done if one moves along paths I and II because these paths begin and end on the same equipotential surface. But, on the other hand, work is done in moving along paths I' and II' and the amount is also the same, because these paths connect the same pair of equipotential surface.



**Fig. 1.** Equipotential surfaces.

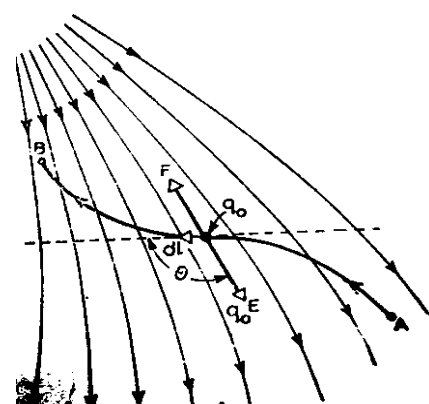


**Fig. 2.** Equipotential surfaces for an isolated positive point charge.

### Electric potential for a non-uniform electric field

Consider a non-uniform field in which the test body is moved along a path which is not *straight*. The electric field exerts a force  $q_0\vec{E}$  on the charge  $q_0$ . In order to move  $q_0$  towards  $B$  an external agent must apply a force  $\vec{F} = -q_0\vec{E}$ . The work done for a small displacement  $d\vec{l}$  along  $AB = \vec{F} \cdot d\vec{l}$ . Therefore, the total work to move test-charge from  $A$  to  $B$  is given by

$$W_{AB} = \int_A^B \vec{F} \cdot d\vec{l} = -q_0 \int_A^B \vec{E} \cdot d\vec{l} \quad (1)$$



**Fig. 3.**

The negative sign is due to the fact that the motion  $A$  to  $B$  is opposite to the direction of field. Hence

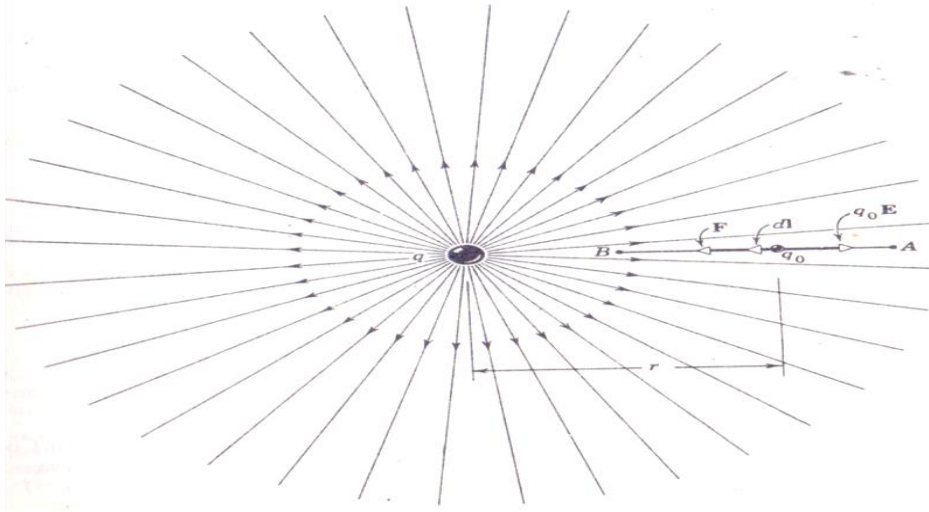
$$V_B - V_A = \frac{W_{AB}}{q_0} = - \int_A^B \vec{E} \cdot d\vec{l} \quad (2)$$

If point  $A$  is situated at infinity and potential  $V_A$  there is zero, then the potential  $V$  at  $B$ .

$$V = - \int_{\infty}^B \vec{E} \cdot d\vec{l} \quad (3)$$

These two equations allow us to calculate the potential difference between any two points (or the potential at any point), if  $E$  is known at various points in the field.

**Electric potential for a point charge:** Fig. 4. shows two points A and B near an isolated point charge  $q$ . For simplicity we assume that A, B, and  $q$  lie on a straight line. Let us calculate the potential difference between the points A and B, assuming that a test charge  $q_0$  is moved along a radial line from A to B.



**Fig. 4.** A test charge  $q_0$  is moved by an external agent from A to B in the field set up by a point charge  $q$ .

As  $\vec{E}$  points in the right direction and test charge is moved towards B i.e., left. Hence  $\vec{E}$  and  $d\vec{l}$  are  $180^\circ$  apart. Therefore,

$$\vec{E} \cdot d\vec{l} = E \cos 180^\circ dl = -Edl$$

Since the distance  $r$  is measured from the charge i.e., right and  $d\vec{l}$  is measured towards B i.e., left. Hence,

$$dl = -dr$$

So the potential difference between the points A and B is given by

$$V_B - V_A = - \int_A^B \vec{E} \cdot d\vec{l} = - \int_A^B -Edl = - \int_A^B E(-dl)$$

$$\therefore V_B - V_A = - \int_{r_A}^{r_B} E dr \quad [dl = -dr]$$

Where  $r_A$  and  $r_B$  are the distances of the points A and B from the point charge.

But,

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$

Therefore,

$$V_B - V_A = - \frac{q}{4\pi\epsilon_0} \int_{r_A}^{r_B} \frac{dr}{r^2} = \frac{q}{4\pi\epsilon_0} \left[ \frac{1}{r_B} - \frac{1}{r_A} \right]$$

If A is situated at infinity and potential there be zero i.e.,  $r_A \rightarrow \infty$  and  $V_A = 0$ , then potential at B if denoted by  $V$ ,

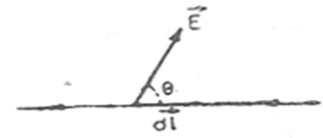
$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$$

The electric potential for a group of point charges is

$$V = \sum_{i=1}^n V_n = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_n}{r_n}$$

### Electric field from potential gradient

If the element of the length  $dl$  is at angle  $\theta$  with respect to the electric field as shown in fig. 5, then the work done for the displacement  $dl$  to move a unit positive charge is the potential change  $dV$ .



**Fig. 5.**

$$dV = \vec{E} \cdot d\vec{l} = -E dl \cos \theta$$

The -ve sign indicates that  $E \cos \theta$  is opposite to the direction of motion, hence

$$\text{grad } V = \frac{dV}{dl} = -E \cos \theta$$

If  $\theta = 0$ ,  $d\vec{l}$  is along the field line, the ratio  $\frac{dV}{dl}$  is maximum, hence

$$\text{gradient of } V = \frac{dV}{dl} = -E$$

Therefore the gradient of the potential (maximum rise in potential with distance) is equal in magnitude to the electric field intensity. *The maximum rise in potential with distance is gradient.* It has direction opposite to the field.

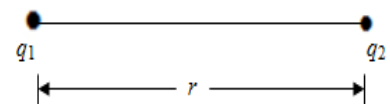
$$\vec{\nabla} V = -\vec{E}$$

$$\therefore \vec{E} = -\vec{\nabla} V$$

The significance of the negative sign is that to experience a rise in potential, a positive test-charge must be moved opposite to the electric field direction.

### Electric potential energy of a system of charges

The electric potential energy of a system of point charges is defined as the work required to assemble this system of charges by bringing them close together, as in the system from an infinite distance.



**Fig. 6**

Let us consider two charges  $q_1$  and  $q_2$  separated by a distance  $r$  as shown in fig. 6. Suppose  $q_2$  is removed to infinity and is at rest. The electric potential at the original site of  $q_2$  caused by  $q_1$  is given by

$$V = \frac{1}{4\pi\epsilon_0} \frac{q_1}{r} \quad (1)$$

Therefore the work required to bring  $q_2$  from infinity to its original position is given by

$$W = Vq_2$$

Substituting the value of  $V$  from equation (1) we get

$$W = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r}$$

Thus, according to the definition of the electric potential energy, this work  $W$  is precisely the electric potential energy  $U$  of the system  $q_1 + q_2$  i.e.,

$$U = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r} \quad (2)$$

### Mathematical problems

**Problem-1:** What must the magnitude of an isolated positive point charge be for the electric potential at 10 cm from the charge to be 100 volts?

**Solution:** We know,

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$$

Or  $q = V \times 4\pi\epsilon_0 r$

Or  $q = 100 \times 4 \times 3.1416 \times 8.854 \times 10^{-12} \times 10$

$\therefore q = 1.11 \times 10^{-9} \text{ C (Ans)}$

Here,

$$V = 100 \text{ volt}$$

$$r = 10 \text{ cm} = 0.10 \text{ m}$$

$$\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2/\text{N} - \text{m}^2$$

$$q = ?$$

**Problem-2:** What is the electric potential at the surface of a gold nucleus? The radius is  $6.6 \times 10^{-15} \text{ m}$  and the atomic number  $Z = 79$ .

**Solution:** We know,

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$$

Or  $V = \frac{9 \times 10^9 \times 79 \times 1.6 \times 10^{-19}}{6.6 \times 10^{-15}}$

$\therefore V = 1.7 \times 10^7 \text{ volt (Ans)}$

Here,

$$q = 79 \times 1.6 \times 10^{-19} \text{ coulomb}$$

$$r = 6.6 \times 10^{-15} \text{ m}$$

$$\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N} - \text{m}^2/\text{C}^2$$

$$V = ?$$

**Problem-3:** What is the potential at the surface of an aluminium nucleus? Given the radius of the nucleus,  $r = 4.5 \times 10^{-13} \text{ cm}$  and its charge  $q = 13e$ . Where  $e$  is the charge of the electron. (Assume that, this charge is concentrated at the center of the aluminium nucleus)

**Solution:** We know,

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$$

Or  $V = \frac{9 \times 10^9 \times 13 \times 1.6 \times 10^{-19}}{4.5 \times 10^{-15}}$

$\therefore V = 4.16 \times 10^6 \text{ volt (Ans)}$

Here,

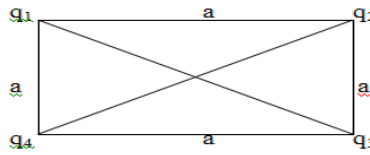
$$q = 13 \times 1.6 \times 10^{-19} \text{ coulomb}$$

$$r = 4.5 \times 10^{-13} \text{ cm} = 4.5 \times 10^{-15} \text{ m}$$

$$\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N} - \text{m}^2/\text{C}^2$$

$$V = ?$$

**Problem-4:** What is the potential at the centre of the following square? Assume that  $q_1 = 1 \times 10^{-8} \text{C}$ ,  $q_2 = -2 \times 10^{-8} \text{C}$ ,  $q_3 = 3 \times 10^{-8} \text{C}$  and  $q_4 = 2 \times 10^{-8} \text{C}$  and  $a = 1.0 \text{ m}$



**Solution:** The distance of each charge from the centre of the square is  $\frac{a}{\sqrt{2}}$  or 0.71 m.

The potential  $V$  at the centre of the square is

$$V = \frac{1}{4\pi\epsilon_0} \sum \frac{q}{r}$$

Or 
$$V = \frac{1}{4\pi\epsilon_0} \left( \frac{q_1}{r} + \frac{q_2}{r} + \frac{q_3}{r} + \frac{q_4}{r} \right)$$

Or 
$$V = \frac{1}{4\pi\epsilon_0} \left( \frac{q_1 + q_2 + q_3 + q_4}{r} \right)$$

Or 
$$V = 9 \times 10^9 \times \left( \frac{1 \times 10^{-8} - 2 \times 10^{-8} + 3 \times 10^{-8} + 2 \times 10^{-8}}{0.71} \right)$$

$\therefore V = 507 \text{ volt (Ans)}$

**Problem-5:** Three charges  $q_1 = -5 \times 10^{-8} \text{Coul}$ ,  $q_2 = 2 \times 10^{-8} \text{Coul}$ ,  $q_3 = 3 \times 10^{-8} \text{Coul}$  are placed respectively at the three corners of an equilateral triangle; the length of any side 'a' of it is 10 cm. Calculate the potential energy of the system.

**Solution:** The potential energy of the system,  $U = U_{12} + U_{23} + U_{31}$

Potential energy between 1st and 2nd charge,  $U_{12} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{12}} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{a}$

$$= \frac{9 \times 10^9 \times (-5 \times 10^{-8} \times 2 \times 10^{-8})}{0.10}$$

$$= -9 \times 10^{-5} \text{ Joules}$$

Potential energy between 2nd and 3rd charge,  $U_{23} = \frac{1}{4\pi\epsilon_0} \frac{q_2 q_3}{r_{23}} = \frac{1}{4\pi\epsilon_0} \frac{q_2 q_3}{a}$

$$= \frac{9 \times 10^9 \times (2 \times 10^{-8} \times 3 \times 10^{-8})}{0.10}$$

$$= 5.4 \times 10^{-5} \text{ Joules}$$

Potential energy between 2nd and 3rd charge,  $U_{31} = \frac{1}{4\pi\epsilon_0} \frac{q_3 q_1}{r_{31}} = \frac{1}{4\pi\epsilon_0} \frac{q_3 q_1}{a}$

$$= \frac{9 \times 10^9 \times (3 \times 10^{-8} \times -5 \times 10^{-8})}{0.10}$$

$$= -13.5 \times 10^{-5} \text{ Joules}$$

The potential energy of the system,  $U = -9 \times 10^{-5} + 5.4 \times 10^{-5} - 13.5 \times 10^{-5}$

$$= -17.1 \times 10^{-5} \text{ Joules (Ans)}$$